

Application of Fuzzy PID Control in UAV Control System

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Abstract— This research studies the application of fuzzy PID control in UAV control system. Based on the modular design concept, in the UAV ground station control system designed in this paper, its structure mainly includes human-computer interaction equipment and UAV instruments, navigation display and flight simulation modules. For the efficient analysis, the fuzzy control model is integrated into the mentioned system. Traditional control methods are widely used in industrial environments, but for systems with large delays and complex nonlinearities, hence, the edge computing and fuzzy clustering is combined as the tool to analyze the comprehensive data and provide the reasonable supporting information. The results show that the designed system is robust.

Keywords— UAV Control; PID Control; System Modelling; Fuzzy Model; Data Mining

I. INTRODUCTION

The UAV subsystem integrates UAV flight platform, measurement digital camera, flight control system integrated with dual frequency GPS, communication system and other equipment. It has the functions of high-precision positioning solution, camera fixed-point exposure, core automatic rotation correction, and obtains high-resolution aerial images to meet the needs of remote sensing image acquisition [1-3]. Early mission planning system functions were relatively single, mainly flight path planning. With the improvement of the autonomous capabilities of drones, the mission planning system gradually developed in the direction of diversification, intelligence, and predictability, making the drone in the mission [4-6].

In the general execution process, complex environmental information can be converted into decision-making control instructions more effectively. In the recent years, lots of the state-of-the-art studies have paid attention to this focus. The existing "ground station remote control drone" mode has defects such as data link interruption and decision delay. It is difficult to support drones to carry out combat missions in a battlefield confrontation environment [7-10].

However, the use of man-machine decision-making mode can achieve countermeasures as the UAV's nearby command. On-site command and measurement and control sub-systems are mainly divided into the protocol conversion module, flight monitoring module, load and the link monitoring, UAV status inspection module, real-time video splicing module and data management module. The on-site command and measurement and control sub-system is deployed on the mobile platform,

which mainly realizes the standardized management of the whole process of UAV field operation tasks in the sea [11-13].

As the reflection, figure 1 denotes the structure of the traditional fuzzy control system.

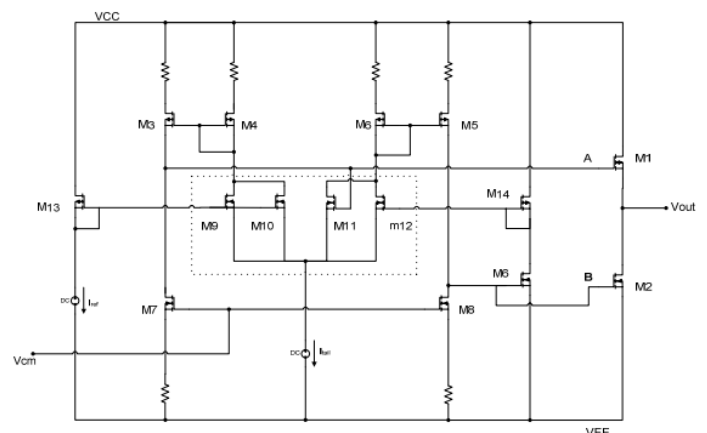


Fig. 1. The Fuzzy Control Model Structure

Traditional control methods are widely used in industrial environments, but for systems with large delays and complex nonlinearities, the instability of some traditional conventional control methods leads to unsatisfactory control effects. Fuzzy control does not require an accurate mathematical model of the controlled object. It is a rule-based control and it is more based on the control experience of the operator. The control amount can be obtained through methods such as table look-up, and the control effect is obvious. Based on this model, the following part of this paper introduces the novelty and the further development trend in detail.

II. THE PROPOSED METHODOLOGY

A. The PID Control Pipelines

Open control system should have standardized interface, modular structure, visual interface and also have extensibility, interoperability, portability and can increase or decrease.

In the stage of stability control of the inverted pendulum, FOPID is used to separately control the position and speed of its pendulum angle and rotation angle. Because there are many controller parameters and numbers, and in the process of the stable control of the inverted pendulum, it is hoped that both the inverted pendulum rod and the rotating rod will obtain a good response at the same time. In the figure 3, we show the core structure of the PID control [14-16].

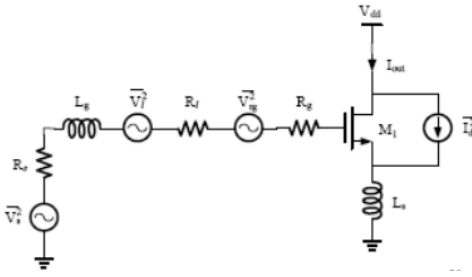


Fig. 3. The Core Structure of the PID Control

The PID controller dynamic equation is as follows:

$$u(t) = PID \left[e(t) + Ti \int_0^t e(t) dt + TD \frac{de(t)}{dt} \right] \quad (1)$$

The designed open control system takes the core industrial computer as the main controller. The external force sensor collects real-time force information and feeds it back to the industrial computer. The force control algorithm processes the force information and calculates the robot pose correction [17].

Fractional calculus is an extension of the theory of integral calculus. The order of calculus is no longer limited to integers, and can be any real number or even a complex number.

Therefore, the transfer function of the fractional PID can be then expressed as:

$$G(s) = k_p + \frac{k_i}{s^\gamma} + k_d s^\alpha \quad (2)$$

We mainly refresh the robot's Cartesian coordinates, joint coordinates and force values collected by force sensors in real-time display to the designated controls and use high-speed drawing control to draw real-time 3D force information.

In general practical applications, the order of the overall adjustment method is proportional, integral, and differential adjustment. At the same time, attention must be paid to the changes in the curve, and the settings and values of all the parameters must be slowly changed to ensure that the product quality conforms to the design. Requirements and standards. In the body adjustment, firstly turn off the differential and integral adjustment, set the integral time to infinite, and the derivative time to zero.

At the very beginning, we can use the core proportional parameter to set it according to relevant experience and values, and then determine the control ratio according to the change of the curve, until the attenuation variable value is one-fourth of the original, then it can be set at this time add to the integral ratio to adjust. Hence, in the formula 3, the updated function with parameter optimization is defined.

$$u_{updated}(k) = u_{old}(k-1) + \Delta u_{old}(k) \quad (3)$$

Parameters in the selection process must abide by the corresponding principles, first, the controlled parameters have a very important decisive factor for the quality of the product. Second, the controlled parameters as far as possible to choose the direct parameters, if the conditions do not allow, then it is necessary to take a corresponding parameter. In the figure 3, the model is defined [18-20].

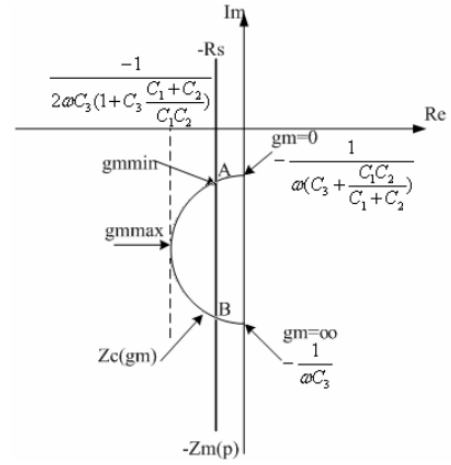


Fig. 3. The PID Control from Different Perspectives

B. The Fuzzy Control

The combination of natural calculation and PID control further improves the control effect of the system, and it has evolved into an intelligent tuning method that integrates a variety of intelligent algorithms.

This article studies the input stage of the SST. The AC-DC circuit uses a single-phase full-bridge PWM rectifier. This circuit can sinusify the input current and also the current and voltage on the grid side are in the same phase to ensure that the rectifier module operates normally under unit power factor with the AC-DC rectification. The traditional variable static pressure control takes the step size as a fixed value in the process of adjusting the static pressure setting value, but it is difficult to determine the appropriate step size: if the step size is too small, the times of adjustment will increase; Over-harmonic oscillation may occur if the step size is too large. In the figure 4, the model structure is defined.

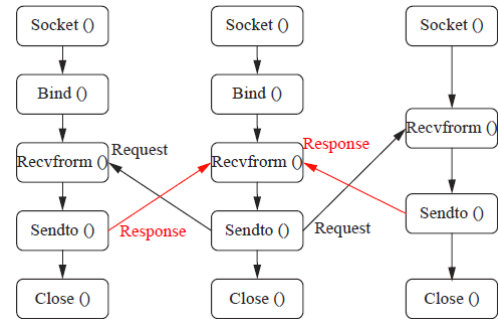


Fig. 4. The Considered Model Structure

The control system of the three-axis stabilized platform is a nonlinear system with strong coupling characteristics.

The three-axis stable platform with the multiple input and output signals has a faster signal change during the movement process, and the coupling effect between the frames of the stable platform will result in low control accuracy of the stable platform. In this paper, the idea of fuzzy control is introduced into the core voltage outer loop control, combined with the traditional PI control, and a fuzzy adaptive control strategy is proposed to enable the system to then adaptively adjust the PI control parameters and improve the steady-state performance

of the system. When the load changes, to reduce the response time of the system and voltage fluctuations. In the formula 4, we define the similarity estimation model [21-23].

$$\frac{\partial y(k)}{\partial u(k)} \approx \sum_{j=1}^L w_j \frac{c_j - x_j}{b_j^2} \quad (4)$$

For the system with strong non-linearity and wide range of operating points, the linear approximation may lead to the instability of the core system. In addition, the descent gradient method for updating and adjusting the parameters of RBF neural network only considers the change of the parameters at the current time, and does not consider the accumulation of experience before and after updating the parameters, which may lead to the oscillation of the learning process and reduce the convergence speed of the network. As the reflection, the figure 5 shows the structure.

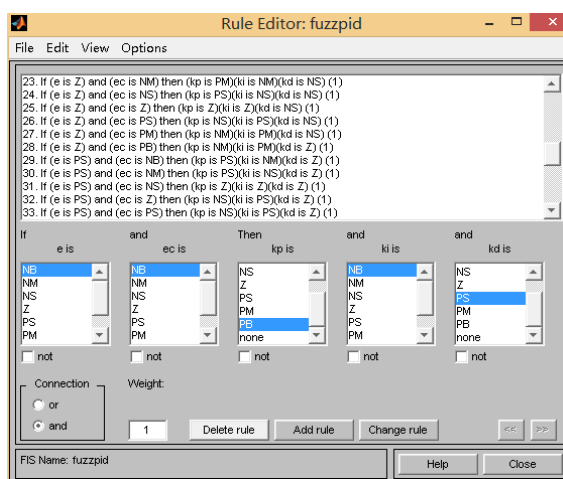


Fig. 5. Fuzzy Control Structure

C. The UAV Control Innovations

The UAV ground station control system can also use the digital radio as the remote measurement data source. After obtaining the corresponding data and performing processing such as noise reduction and filtering, it can be fed back to the corresponding display module.

According to the flying principle of the UAV and its core control requirements, it can be seen that in the UAV ground station control system, the core remote control commands are mainly composed of switches, proportional commands, and uplink navigation data. The mission planning system, as the carrier of the UAV test flight process and test flight tasks, designs the routes, target areas, tactical actions, and mission load application methods that the UAV needs to complete the designated tasks, and adopts information on the specific flight tasks and requirements. The method of the transformation is transformed into a data structure that can be recognized and executed by the drone. In summary, we consider 3 aspects.

(1) The vehicle-mounted positioning system is composed of Beidou navigation and positioning terminals, which can realize the navigation and positioning of emergency vehicles, and can upload position information to the command center in

real time in the form of short messages to realize the real-time monitoring and positioning function of the mobile platform.

(2) The integrated support system is well composed of the intelligent power distribution and central control, intelligent handheld terminals, walkie-talkies, comprehensive monitoring of vehicle equipment and other equipment. It can realize the intelligent control of the equipment power supply, equipment operation, and real-time monitoring of vehicle power supply and equipment running status.

The functional configuration and also the task application configuration of different UAV autonomous control systems make the UAV have different autonomous control capabilities to adapt to different autonomous requirements and complete different tasks. Its main function is to perceive, plan, make decisions and act the system, etc. In the figure 6, we denote the core data collection results.

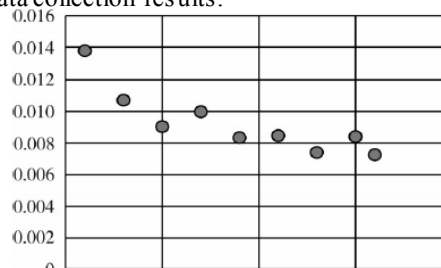


Fig. 6. Fuzzy UAV Control Data Collection Result

III. PERFORMANCE EVALUATION

In this section, the simulation is conducted. We added the fuzzy PID control algorithm to the control board program of the micro three-axis stabilized platform, and compared the performance with the classic PID control algorithm on the micro three-axis stabilized platform. In the figure 7, we firstly show the simulation on the robustness.

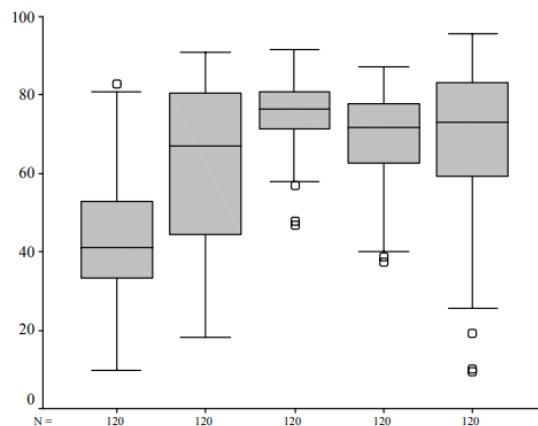


Fig. 7 Fuzzy Result on Robustness Testing

Usually evaluate the pros and cons of a control algorithm, mainly by looking at responsiveness, stability and accuracy of the system under the action of the algorithm. The performance evaluation index is used to measure the quality of the control system algorithm. Therefore, by choosing an objective and some reasonable system performance evaluation index is more

important for judging the effect of the control algorithm. The figure 8 shows the estimated result.

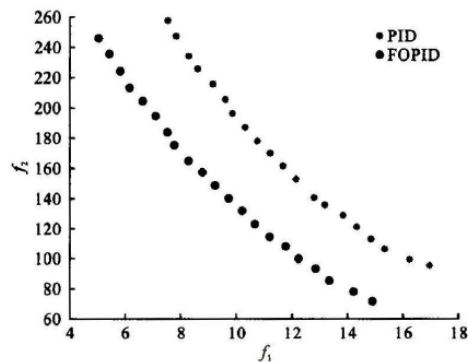


Fig. 8. Comprehensive Simulation on the Results

IV. CONCLUSION

This research studies the application of fuzzy PID control in UAV control system. Based on the background of manned-aircraft/UAV formation cooperative combat and air attack operations, we verify the logical rationality of the architecture of the manned-aircraft/UAV formation cooperative combat decision-making system. We combine the analysis of the core fuzzy model and the core general framework to construct the efficient model. The experimental results show that the model is efficient and the results are satisfactory.

V. ACKNOWLEDGEMENT

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